

# Aryan Singh

Machine Learning Engineer | Data Scientist

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## PROFILE SUMMARY

**Machine Learning Engineer** and **Data Scientist** with practical experience in **Machine Learning**, **Deep Learning**, **Data Analysis**, and **Flask Application Development**. Proficient in **Python**, **SQL**, and **Predictive Modeling**, with expertise in developing data-driven solutions and scalable applications. Skilled in building end-to-end AI solutions, from data analysis and model training to deployment using **Flask/FastAPI**. Passionate about leveraging **Artificial Intelligence**, **Data Science**, and **Software Engineering** to deliver impactful solutions.

## SKILLS

Python, SQL (MySQL), Tableau

Data Cleaning, Data Analysis, Data Visualization, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering, Data Preprocessing, Model Training, Model Evaluation, Hyperparameter Tuning, Classification, Regression, Neural Networks, Machine Learning, Deep Learning, PyTorch, Scikit-learn, NLP (spaCy), Flask, FastAPI, Git, GitHub

## INTERNSHIP EXPERIENCE

**Certify Technology**

**Nov 2025 - Dec 2025**

**Web Developer intern**

**Remote, India**

- Developed and maintained **backend web applications** using **Python**, **Flask**, and **SQL databases**.
- Designed and integrated **RESTful APIs** to enable secure and efficient communication between application components.
- Implemented database operations, data validation, and business logic to support scalable web solutions.
- Optimized application performance and collaborated with cross-functional teams to deliver reliable and user-centric software.

**Rogue Code**

**July 2025 - July 2025**

**DS & DL intern**

**Mumbai, India**

- Designed and implemented end-to-end **Machine Learning** and **Deep Learning** workflows, covering **data preprocessing**, **feature engineering**, and **model evaluation**.
- Built and trained **PyTorch-based neural networks**, including **CNN** and **LSTM** architectures, for classification and prediction tasks.
- Conducted model validation, performance analysis, and **hyperparameter tuning** to enhance accuracy and generalization.
- Utilized **Python**, **PyTorch**, and **Scikit-learn** to develop scalable AI solutions for real-world applications.

## PROJECTS

**Swar-Samved: AI Music Tutor**

**May 2026**

**Python | FastAPI | Librosa | React | Next.js | TypeScript**

**Personal Project**

- Developed an **AI-powered vocal analysis platform** that evaluates **pitch**, **rhythm**, and **tonal accuracy** in real time using advanced audio signal processing techniques.
- Implemented **audio feature extraction**, **spectrogram analysis**, and **PYIN-based pitch detection** using **Librosa**, **NumPy**, and **SciPy** to generate intelligent performance insights.
- Built a full-stack application with **FastAPI**, **React**, **Next.js**, and **Web Audio API** for real-time audio recording, visualization, and personalized feedback generation.

**Human Activity Recognition System**

**July 2025**

**Python | PyTorch | LSTM | Deep Learning**

**Internship Project**

- Developed a **Human Activity Recognition (HAR)** system using **LSTM-based Deep Learning** to classify human activities from time-series sensor data with high accuracy.
- Performed **data preprocessing**, **feature engineering**, and sequence generation, followed by model training and evaluation using PyTorch.
- Evaluated model performance using **accuracy**, **precision**, **recall**, and **confusion matrix metrics**, enabling efficient real-time activity prediction.

## EDUCATION

**Vidyavardhini's College of Engineering and Technology**

**2023-2027**

**B.E. Computer Science Engineering (Data Science)**

**Vasai, Palghar**

## CERTIFICATIONS

- Coursera - IBM Data Science Professional Certificate
- Coursera - IBM Machine Learning Professional Certificate